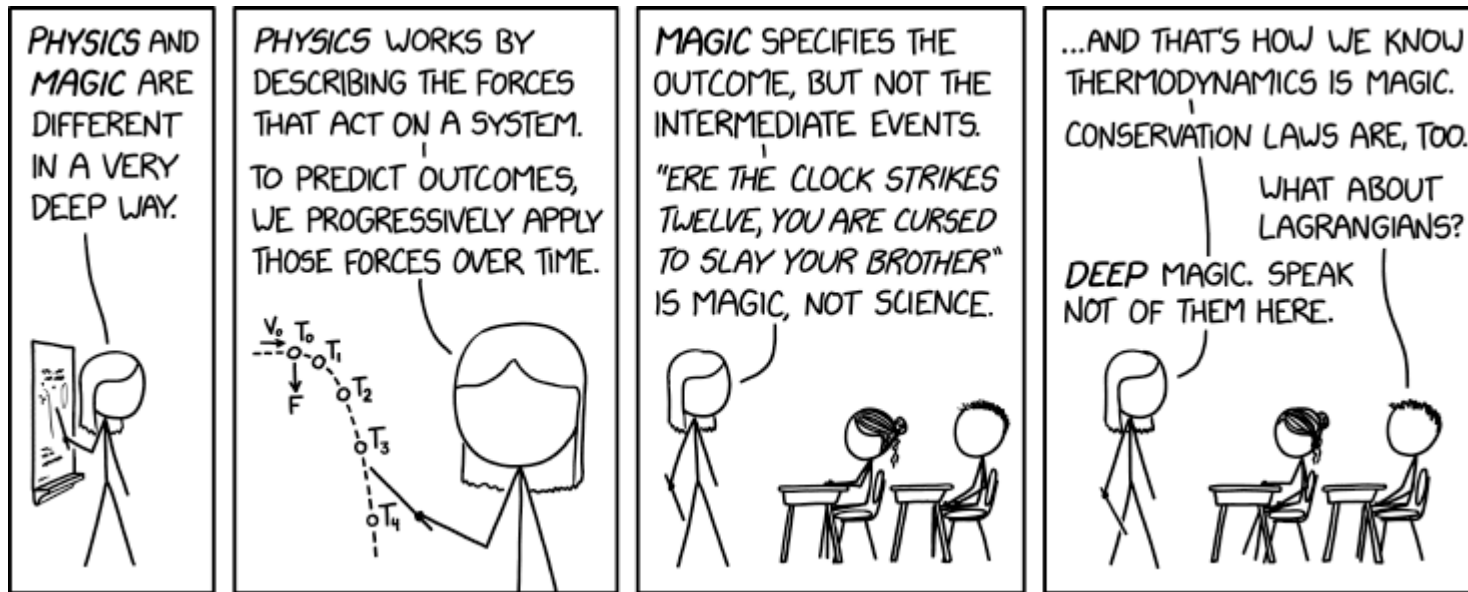




<https://xkcd.com/2904/>

Classic Mechs EP4: Sinks and Saddles and Dynamics

Classic Mechs EP4: “Eigenstuff” for dynamics

Learning objectives

- 1) Review the “Eigenstuff” we did last week for system of equations
- 2) Check out some dynamic systems in phase portrait
- 3) A little intro into Lagrange mech, where we can derive equations of motions but “ignoring” all the forces

Last week: diagonal form for the general solution

For general system $\dot{\underline{x}} = \underline{\underline{A}} \underline{x}$

$$\underline{\underline{A}} \underline{\underline{T}} = \underline{\underline{T}} \underline{\underline{D}}$$

Eigenvalues already!

T is made with columns of Eigenvectors

$$\underline{x}(t) = \underline{\underline{T}} e^{\underline{\underline{D}}t} \underline{\underline{T}}^{-1} \underline{x}(0)$$

x Single underline: Vector

A Double underline: Matrix

Two things before we continue:

1) Why this:

$$\underline{x}(t) = \underline{T} e^{\underline{D}t} \underline{T}^{-1} \underline{x}(0)$$

If we can:

$$\underline{x}(t) = e^{\underline{A}t} \underline{x}(0)$$

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Answer: Matrix exponential don't work like $\underline{D}$ matrix for $\underline{A}$!

Diagonalizable case [[edit](#)]

If a matrix is [diagonal](#):

$$A = \begin{bmatrix} a_1 & 0 & \cdots & 0 \\ 0 & a_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a_n \end{bmatrix},$$

then its exponential can be obtained by exponentiating each entry on the main diagonal:

$$e^A = \begin{bmatrix} e^{a_1} & 0 & \cdots & 0 \\ 0 & e^{a_2} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & e^{a_n} \end{bmatrix}.$$

https://en.wikipedia.org/wiki/Matrix_exponential

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1) Why this:

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If we can:

$$\underline{x}(t) = e^{\underline{A}t} \underline{x}(0)$$

Answer: Matrix exponential don't work like D matrix for A!

2) Solving eigenvector: $\underline{A} \underline{t}_n = \lambda_n \underline{t}_n$

For specific eigenvalue λ_n , there're a corresponding direction of eigenvector $\underline{t}_n$!

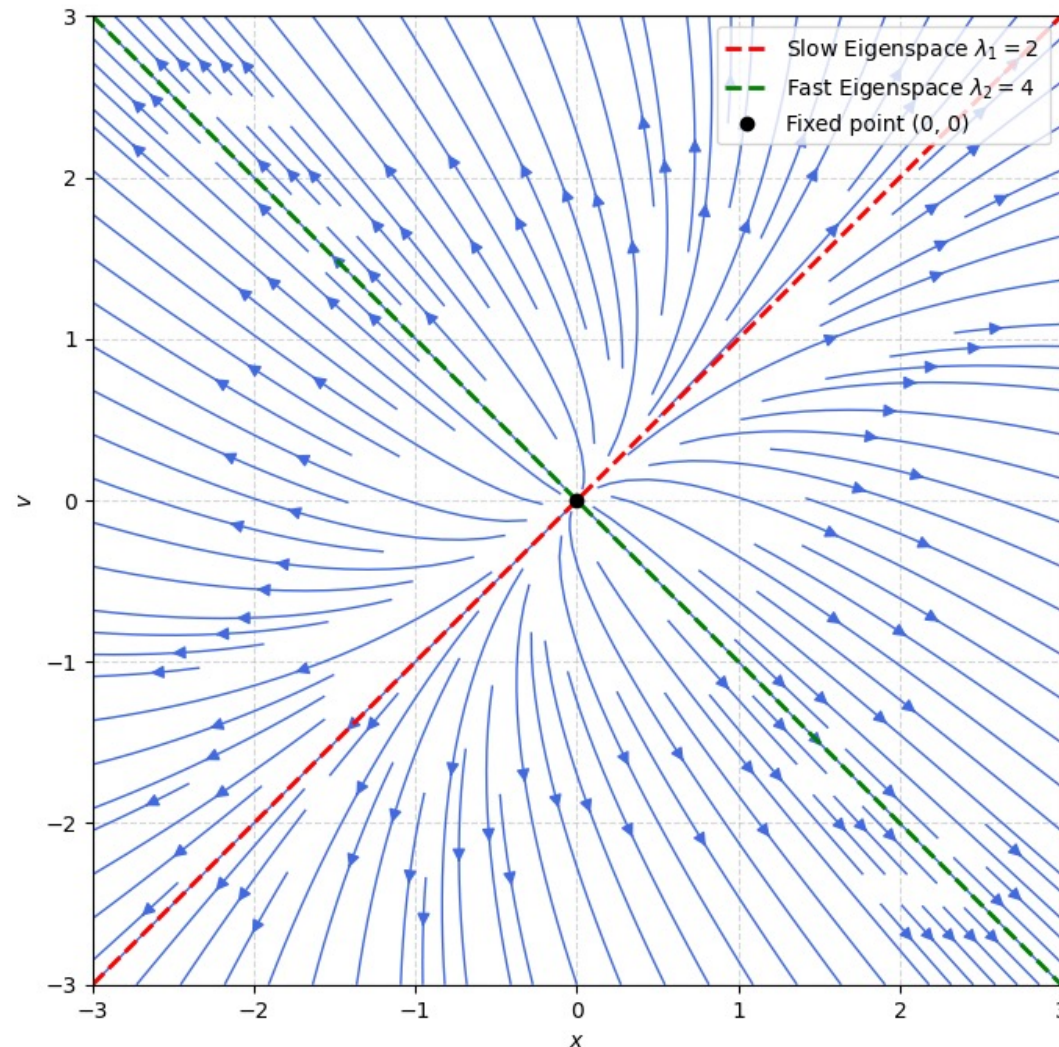
Example of unstable source:

$$\dot{\underline{x}} = \underline{A} \underline{x}$$

$$\underline{A} = \begin{bmatrix} 3 & -1 \\ -1 & 3 \end{bmatrix}$$

$$\underline{T} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

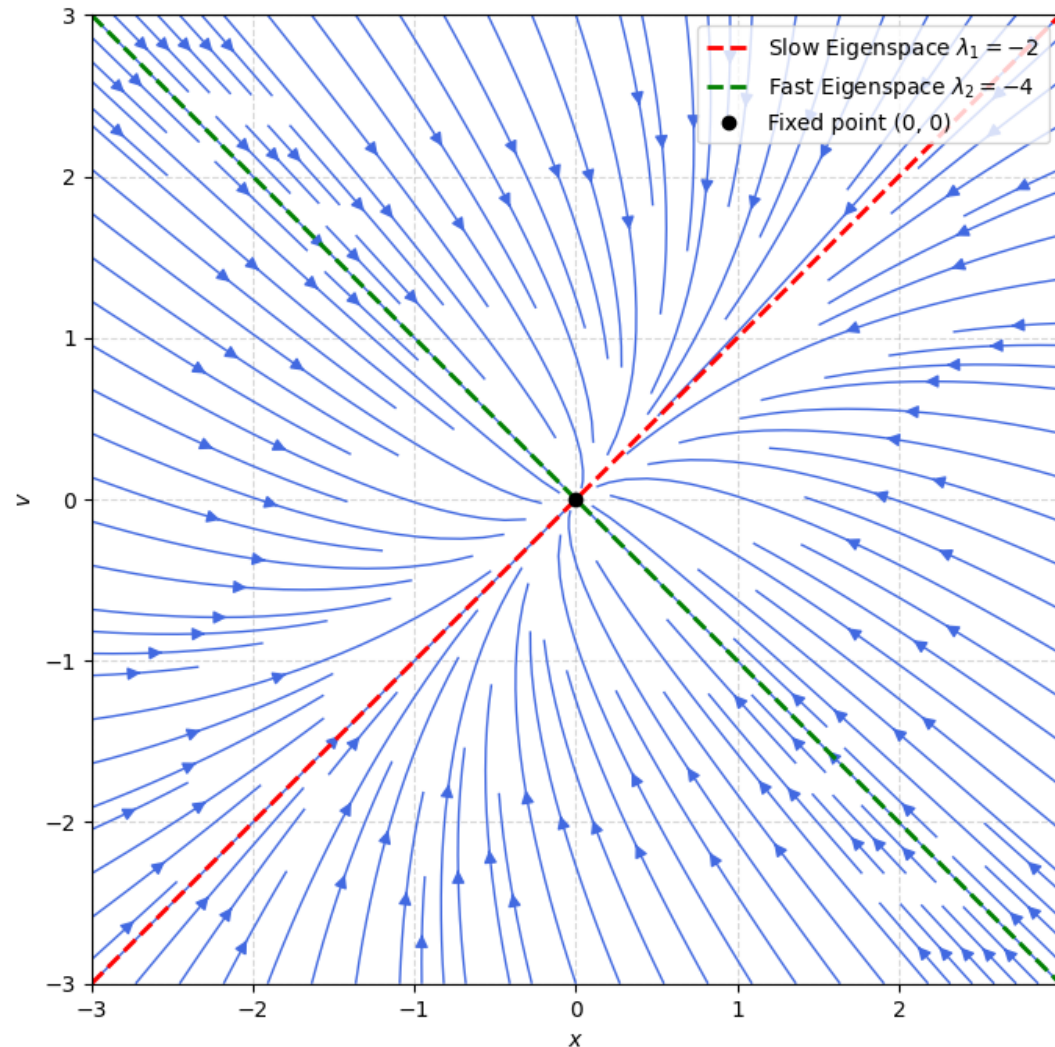
$$\underline{D} = \begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix}$$



Example of stable sink:

$$\dot{\underline{x}} = \underline{A} \underline{x}$$

$$\underline{A} = - \begin{bmatrix} 3 & -1 \\ -1 & 3 \end{bmatrix}$$



Example of saddle point:

$$\dot{\underline{x}} = \underline{A} \underline{x}$$

$$\underline{A} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

Example of saddle point:

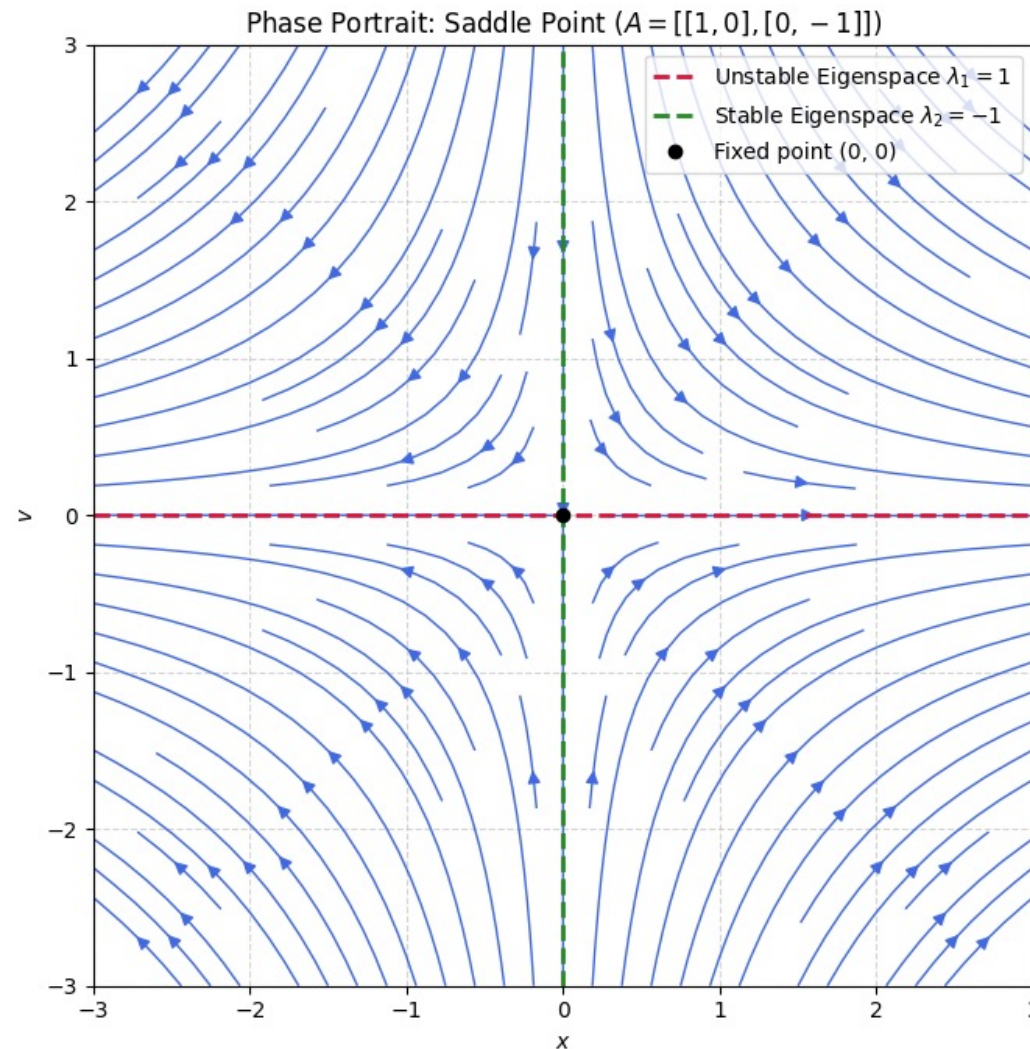
$$\dot{\underline{x}} = \underline{A} \underline{x}$$

$$\underline{A} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

Example of saddle point:

$$\dot{\underline{x}} = \underline{A} \underline{x}$$

$$\underline{A} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$



Example of ?

$$\underline{\dot{x}} = \underline{A} \underline{x}$$

$$\underline{A} = \begin{bmatrix} 0 & -2 \\ 2 & 0 \end{bmatrix}$$